Soren Filippi

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Aerospace engineering MSc student at ISAE-SUPAERO, aiming for propulsion. Experience modelling and controlling a real engine on a test bench, now testing a hydrogen fuel cell for a hybrid-electric aircraft.

EDUCATION

ISAE-SUPAERO, Toulouse, France — 2026 – present
MSc in Aerospace Engineering (MAE, first year)

ETH Zürich (Swiss Federal Institute of Technology), Zürich, Switzerland — 2025 – 2026
Exchange year in Mechanical Engineering

- 14 master-level courses across five departments (63 ECTS), concentrated in aeronautics and aerospace
- Engine Systems, Combustion, Compressible Flows, Control Systems II, Finite Element Analysis, Machine Learning

EPFL (Swiss Federal Institute of Technology), Lausanne, Switzerland — 2022 – 2026
BSc in Mechanical Engineering

- 95th percentile in MàN; strong foundation in mathematical modelling and data analysis

Lycée Français Charles de Gaulle, London, UK — 2019 – 2022

- French Baccalaureate: 17/20 overall, 18/20 in Physics-Chemistry and Maths, 20/20 in Expert Maths
- Class representative for 7 years, member of the Disciplinary Committee, honours student

ENGINEERING PROJECTS

Idle speed control of a 2.5 L five-cylinder engine, ETH Zürich — Team lead, team of 4

- Identified a mean-value engine model in Simulink from test bench data; designed an LQG controller with integral action that uses ignition timing for fast disturbance rejection and the throttle for steady load
- Tuned the controller by constrained optimisation in MATLAB against the competition cost, with robustness constraints; re-tuned on data from the first bench session
- On the real engine: steady-state speed error below 0.5 rad/s, back on target within 2.4 s after a 1 kW load step

Reliability of SWOT satellite water levels over Swiss lakes, ETH Zürich — Team of 6

- Led the initial strategy and the data analysis for a lake-independent Python pipeline processing NASA/CNES SWOT data
- Validated against federal gauges on five lakes: centimetre-level mean offsets, correlation 0.88 to 0.99; co-wrote the research report

Hydrogen fuel cell testing, HydroArt, ISAE-SUPAERO student association — 2026 – present

- Fuel cell team for a 1/8 scale aircraft replica with hybrid fuel cell and battery propulsion; preparing a test campaign on an 800 W PEM stack

Numerical methods coursework, ETH Zürich — Individual

- Wrote finite element solvers from scratch in Python: trusses, 2D frames, CST and Q4 elements with Gauss quadrature, explicit and implicit finite differences
- Built a U-Net in PyTorch to reconstruct masked image regions, about 3.5 times lower error than a mean-fill baseline

Hand-cranked winch for felled trees, EPFL — Group leader, team of 4

- Led the design from client specification to a full CATIA model and video presentation; did most of the sizing calculations

EXPERIENCE

1751 Data, London, UK — Jul – Aug 2025
Intern

- Designed tear-sheet templates that summarise how well market-prediction models perform against a reference benchmark, using key performance indicators
- Built a Python data processing tool that turned large, unstructured datasets into these client-ready visual summaries

Polanime (student association), Lausanne, Switzerland — 2023 – 2025
Event organiser and staff volunteer

- Coordinated logistics and staff for large campus events, including a ski weekend for 250 students

SKILLS & INTERESTS

Languages	French (native), English (fluent), Spanish (business)
Technical	Python (NumPy, SciPy, PyTorch, scikit-learn), MATLAB/Simulink, C/C++, LabVIEW, CATIA
Interests	Pottery (4 years), MAKE projects at EPFL, UNIL running club, Duke of Edinburgh Silver